CENFLOX 100 mg/ml
solution for injection

STATEMENT OF THE ACTIVE SUBSTANCES AND OTHER INGREDIENTS
Each ml contains:
Active substance:
Enrofloxacin..100.0 mg
Excipients:
n-Butanol ....30.0 mg (as preservative)
Benzyl alcohol (E1519)...........20.0 mg (as preservative)

Clear, yellow solution

INDICATIONS
Cattle: For the treatment of respiratory tract infections caused by enrofloxacin-sensitive *Histophilus somni*, *Mannheimia haemolytica, Pasteurella multocida* and *Mycoplasma* spp. sensitive to enrofloxacin, as well as for the treatment of colimastitis.
Pigs: Treatment of bacterial bronchopneumonia caused by *Actinobacillus pleuropneumoniae*, *Haemophilus parasuis* and *Pasteurella multocida* sensitive to enrofloxacin.

CONTRAINDICATIONS
Do not use in cases of hypersensitivity to the active substance or to any of the excipient. Do not use in animals with central nervous system-associated seizure disorders. Do not use in the presence of existing disorders of cartilage development or musculoskeletal damage around functionally significant or weight-bearing joints.
Do not use in case of resistance against other fluoroquinolone due to the potential for cross-resistance against them.

SIDE EFFECTS
On very rare occasions the following can occur:
- Transitory inflammatory reactions (swelling, redness) can occur at the injection site in very rare cases. These regress within a few days without further therapeutic measures.
- Shock reactions after intravenous injection in cattle, probably as a result of a circulatory problem.
- Gastrointestinal disorders during treatment of calves.
The frequency of adverse reactions is defined using the following convention:
- very common (more than 1 in 10 animals treated displaying adverse reaction(s))
- common (more than 1 but less than 10 animals in 100 animals treated)
- uncommon (more than 1 but less than 10 animals in 1,000 animals treated)
- rare (more than 1 but less than 10 animals in 10,000 animals treated)
- very rare (less than 1 animal in 10,000 animals treated, including isolated reports).

TARGET SPECIES
Cattle and pigs

DOSAGE FOR EACH SPECIES, ROUTES AND METHOD OF ADMINISTRATION
Cattle:
The dosage for respiratory disease is 7.5 mg enrofloxacin per kg body weight (bw) for a single treatment by subcutaneous administration (s.c.). This is equivalent to 7.5 ml of the product per 100 kg bw and day. Do not administer more than 15 ml (cattle) or 7.5 ml (calf) per injection site (subcutaneous). In case of serious or chronic respiratory disease a second injection may be required after 48 hours.
The dosage for the treatment of colimastitis is 5 mg enrofloxacin per kg body weight (bw) by intravenous administration (i.v.). This is equivalent to 5 ml of the product per 100 kg bw and day. Treatment of colimastitis is exclusively intravenous for 2 to 3 consecutive days.
Pigs:
The dose for respiratory diseases is 7.5 mg of enrofloxacin per kg of live weight for a single treatment. This is equivalent to 0.75 ml of veterinary medicinal product per 10 kg of l.w. per day. Do not administer more than 7.5 ml at the same injection site (intramuscular). In case of severe or chronic respiratory diseases, a second injection may be necessary 48 hours later.

ADVICE ON CORRECT ADMINISTRATION
Cattle: Subcutaneous injection (respiratory disease) or intravenous injection (colimastitis).
Pigs: Intramuscular injection into the neck musculature, behind the ear.
The animal weight must be determined as accurately as possible to avoid insufficient dosage.
The vial cap can be safely pierced up to 30 times.

WITHDRAWAL PERIOD
Cattle:
Meat: s.c.: 14 days, i.v.: 7 days.
Milk: s.c.: 120 hours (5 days), i.v.: 72 hours (3 days).
Pigs:
Meat: i.m.: 12 days.

SPECIAL STORAGE PRECAUTIONS
Shelf-life after first opening the immediate packaging: 28 days.
The storage conditions once package is opened are: Keep the container in the outer carton. Do not store above 30ºC.

SPECIAL WARNINGS
Special precautions for use in animals:
When using this medication, the official recommendations on the use of antimicrobials should be taken into account. The fluoroquinolones should be reserved for the treatment of infections that have not responded or are expected not to respond correctly to other groups of antimicrobials. Whenever possible, the use of fluoroquinolones should be based on the sensitivity tests. The use of the medicine in different conditions from those recommended in the Data Sheet can increase the prevalence of bacteria resistant to the fluoroquinolones and decrease the effectiveness of the treatment with other quinolones as a consequence of the appearance of cross-resistances. A new injection site should be selected when more than one injections is given or if the volume to be administered is greater than 15 ml (cattle) or 7.5 ml (calves, pigs). Enrofloxacin is excreted through the kidneys. As with all other fluoroquinolones, excretion in animals with kidney lesions may be delayed.
Special precautions to be taken by the person administering the veterinary medicinal product to animals:
Avoid direct skin contact due to sensitisation, contact dermatitis and possible hypersensitivity reactions. People with known hypersensitivity to (fluoro)quinolones should avoid contact with the product. Wash hands after use.
In case of accidental splashes in the eyes, wash with plenty of clean water.
Do not eat, smoke or drink while handling the medicinal product.
Precautions should be taken to avoid accidental self-injection.
In case of accidental self-injection, seek medical advice immediately and show the package leaflet or label to the physician.
Use during pregnancy, lactation or lay:
It can be used during pregnancy and lactation.
Interaction with other medicinal products and other forms of interaction:
Antagonist effects may occur in conjunction with macrolides or tetracyclines.
Enrofloxacin may interfere with the metabolism of theophylline, decreasing theophylline clearance resulting in increased plasma levels of theophylline.

Overdose (symptoms, emergency procedures, antidotes), if necessary:

In cattle a dose of 25 mg/kg of live weight was tolerated subcutaneously for 15 consecutive days without the appearance of clinical symptoms. Higher doses in cattle and doses of about 25 mg/kg and higher in pigs can cause lethargy, lameness, ataxia, mild salivation and muscle tremors.

Do not exceed the recommended dose. In accidental overdose there is no antidote and treatment should be symptomatic.

SPECIAL PRECAUTIONS FOR THE DISPOSAL OF UNUSED PRODUCT OR WASTE MATERIALS, IF ANY

Any unused veterinary medicinal product or waste materials derived from such veterinary medicinal product should be disposed of in accordance with local regulation.

OTHER INFORMATION

Pharmacodynamic properties

Enrofloxacin is an antimicrobial belonging to the fluoroquinolones group. The substance performs bactericidal activity on DNA-gyrase and topoisomerase IV, selectively inhibiting these enzymes. DNA-gyrase and topoisomerase IV are type II topoisomerases present in bacteria. These enzymes are involved in the replication, transcription and recombination of bacterial DNA. Fluoroquinolones also act on the bacteria in the stationary phase by altering the permeability of the bacteria's cell wall. Inhibitory and bactericidal concentrations of enrofloxacin are very similar. Either they are the same, or at most, they differ in 1-2 dilution steps. Enrofloxacin has an action spectrum that includes the enrofloxacin-sensitive bacteria *Histophilus somni*, *Manheimia haemolytica*, *Pasteurella multocida*, *Mycoplasma spp*. and *Escherichia coli* in cattle, as well as *Actinobacillus pleutopneumoniae*, *Pasteurella multocida* and *Haemophilus parasuis* in pigs. Five mechanisms of resistance to fluoroquinolones have been described (i) point mutation of the genes encoding the DNA-gyrase and/or topoisomerase IV causing changes in the respective enzymes, (ii) changes in drug permeability in Gram-negative bacteria, (iii) expulsion mechanisms, (iv) plasmid-mediated resistance and (v) protective gyrase proteins. These mechanisms cause bacteria to be less sensitive to fluoroquinolones. Cross-resistance between fluoroquinolone class antimicrobials is common. Enrofloxacin clinical cut-off points (Susceptible, Intermediate, Resistant) are available for:

Strains of *Mannheimia haemolytica*, *Pasteurella multocida* and *Histophilus somni* isolated from cattle (S< 0.25 µg/ml; I=0.5 µg/ml, R > 2 µg/ml, CLSIVET08ED4-2018), strains of *Pasteurella multocida* and *Actinobacillus pleuropneumoniae* isolated from pigs (S< 0.25 µg/ml; I=0.5 µg/ml, R > 1 µg/ml, CLSIVET08ED4-2018). There are no clinical cut-off points available for *E.coli* strains isolated from cattle/mastitis (ECOFF= 0.125 µg/ml, EUCAST 2019).

CMI90 values have been reported for *E.coli* strains isolated from clinical mastitis of 0.06 – 0.125 µg/ml (Sweden: 0.125 µg/ml 2013-2017, 503 strains; Czech Republic: 0.125 µg/ml 2013-2017, 192 strains; Denmark: 0.06 µg/ml 2004-2014, 1756 strains).

Pharmacokinetic properties

After administration of the drug subcutaneously in cattle and intramuscularly in pigs absorption of the active substance enrofloxacin is very rapid and almost complete (high bioavailability).

Cattle:

After subcutaneous administration of 7.5 mg of enrofloxacin/kg of live weight in non-lactating cattle, the maximum plasma concentration of 0.82 mg/L is reached within 5 hours. The total plasma exposure to the substance is 9.1 mg h/L. Enrofloxacin leaves the body with a half-life of 6.4 hours. Approximately 50% of enrofloxacin is metabolised to the active substance ciprofloxacin. Ciprofloxacin leaves the body with a half-life of 6.8 hours.

After an intravenous injection of 5.0 mg of enrofloxacin/kg of live weight into lactating cows, the maximum plasma concentration of approximately 23 mg/L is reached immediately. The total plasma exposure of the substance is 4.4 mg h/L. Enrofloxacin leaves the body with a half-life of 0.9 h. Approximately 50% of enrofloxacin is metabolised to the active substance ciprofloxacin. Ciprofloxacin leaves the body with a half-life of 6.8 hours.

After an intravenous injection of 5.0 mg of enrofloxacin/kg of live weight into lactating cows, the maximum plasma concentration of approximately 23 mg/L is reached immediately. The total plasma exposure of the substance is 4.4 mg h/L. Enrofloxacin leaves the body with a half-life of 0.9 h. Approximately 50% of the original compound is metabolised to ciprofloxacin with maximum plasma concentrations of 1.2 mg/L, which are reached at 0.2 h. The average elimination half-life of ciprofloxacin is 2.1 h.

In milk, the metabolite ciprofloxacin has the main responsibility for antibacterial activity (approx. 90%). Ciprofloxacin reaches maximum milk concentrations of 4 mg/L within 2 hours after intravenous administration. The total exposure in milk after 24 hours is approximately 21 mg h/L. Ciprofloxacin leaves the milk with a half-life of 2.4 h. In milk, maximum concentrations of 1.2 mg of enrofloxacin per litre are reached in the next 0.5 hour, with a total exposure of enrofloxacin in milk of approximately 2.2 mg h/L. Enrofloxacin is excreted from milk at 0.9 h.

Pigs:

After the intramuscular administration of 7.5 mg/kg of l.w. in pigs, the mean maximum serum concentration of 1.46 mg/L was reached in the next 4 hours. Total 24-hour exposure to the substance was 20.9 mg h/L. The substance was excreted from the central compartment with a terminal half-life of 13.1 h. With maximum concentrations lower than 0.06 mg/L, mean serum concentrations of ciprofloxacin were very low.

Enrofloxacin has a high distribution volume. Concentrations in tissues and organs often significantly exceed plasma levels. The organs where a high concentration can be expected are the lungs, liver, kidney, intestine and muscle tissue. Enrofloxacin is excreted through the kidneys.

INCOMPATIBILITIES

In the absence of compatibility studies, this veterinary medicinal product should not be mixed with other veterinary medicines.

Pack sizes: Box containing 1 vial of 100 ml. Box containing 1 vial of 250 ml. Box containing 10 vials of 100 ml. Box containing 10 vials of 250 ml.

Not all pack sizes may be marketed.

For animal treatment only. To be supplied only on veterinary prescription.

Keep out of the sight and reach of children.

"Jauhi ubat dari kanak-kanak"

Ubat Terkawal

DATE OF REVISION OF LEAFLET

January 2022

MARKETING AUTHORISATION HOLDER

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